HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Java Spring Boot** microservices for backend systems, ensuring seamless data exchange through **REST APIs**.
- Developed and maintained backend services using **Hibernate** and **JPA** for efficient database interaction and data persistence.
- Enhanced application performance and reliability by writing **JUnit** tests and employing **Mockito** for mocking dependencies in **Java EE** environments.
- Collaborated with cross-functional teams to define and refine technical requirements for new manufacturing process equipment, contributing to sprint planning and technical estimations.
- Authored and updated comprehensive **troubleshooting documentation** and training materials for production equipment, enhancing operational knowledge transfer.
- Applied a **scientific approach** to problem-solving for equipment and process improvements, blending analytical and experimental skills to maximise productivity and yields.
- Containerized backend applications using **Docker** and deployed them to **AWS EC2** instances, managing configurations and basic resource allocation.

SKILLS

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, Hibernate, JPA, JUnit, Mockito, pandas, NumPy, scikit-learn, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Platforms – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau

Methodologies – Agile, Scrum, CI/CD

PROJECTS

Java Backend for Automated Manufacturing Monitoring

- Developed a **Java Spring Boot** application to ingest real-time data from automated manufacturing equipment, processing ~5,000 data points per minute.
- Implemented **REST APIs** for equipment status reporting and alert notification, integrating with a central monitoring dashboard.
- Utilized **PostgreSQL** for storing historical performance data and implemented efficient query patterns for reporting analysis.

Python Data Pipeline for Production Yield Analysis

- Engineered a **Python** data pipeline using **pandas** to extract, transform, and load production data from various sources, including CSV files and database extracts.
- Developed automated scripts to calculate key yield metrics and generate daily reports, identifying process deviations affecting output by up to **15%**.
- Deployed the pipeline on **AWS Lambda** for scheduled execution, ensuring timely delivery of analytical insights to the operations team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)